

AI 507: Artificial Intelligence and Society



#4 Communication

Associate Prof. Dr. Lena Frischlich

Flashlight

Flashlight

Social cognition

Humans' default setting

Self concept

Working self-concept

Situational self-concept

Self-schemes

Possible self

Feared self

Self regulation

What Self-discrepancy

When Self-efficacy

Why Central needs

Accuracy

Self-enhancement

Consistency

AI-self?

Interpreting others

Humans' default setting

Biases

Fundamental attribution error

Naïve realism

Synchopathic AI

Social groups

Social Identity / Self-categorization theory

Ingroups

Outgroups

Stereotype content model

Competence

Warmth

Debiasing humans

Counterexemplars

Intergroup contact hypothesis

Debiasing AI?

Any questions to last week?



Then let's get started!

After today's lecture, I hope you will...

- ...have an initial understanding of communication
- ...have examined the production and processing of communication
- ...have gotten a glimpse into the complex processes behind reading
- ...explored how cultures and computers shape communication

What is communication?

Think for a moment about the role of communication in your life

(Krcmar et al., 2016)

- How much time of the day do you spend communicating?
- Does communication make your life more enjoyable?
- Does communication change how you see yourself and the world?
- How does communication impact your relationships?
- When did you start communicating?



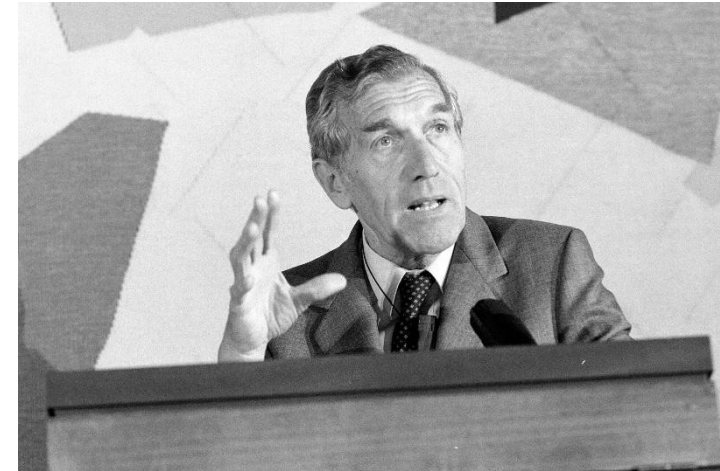
M.C. Escher, *Bond of Union*, April 1956, lithograph. The M.C. Escher Collection, Gemeentemuseum Den Haag, The Hague, the Netherlands © The M.C. Escher Company

Communication is central!

But what *is* communication?

(Sylvia IV; 2024; Watzlawick et al., 1967, CH2, p. 13)

- **Inevitable:** one cannot *not* communicate
- **Episodic:** Describing when communication starts is like an attempt to photograph dancing: we typically make a somewhat arbitrary cut in analysing communication episodes
- **Constructed:** Communication emerges from the meaning assigned verbal & non-verbal symbols and signs – both by the sender and the recipient
- **Contextual:** The interpretation of communication is influenced by multiple contexts – both within each communication and in the larger cultural context



Paul Watzlawick - Von Universitätsarchiv St.Gallen |
Regina Kühne | HSGH 022/000762 | CC-BY-SA 4.0, CC
BY-SA 4.0,
<https://commons.wikimedia.org/w/index.php?curid=139121260>



<https://www.pickpik.com/ballet-dance-ballerina-scene-dancer-romance-1291>

Communication is mediated

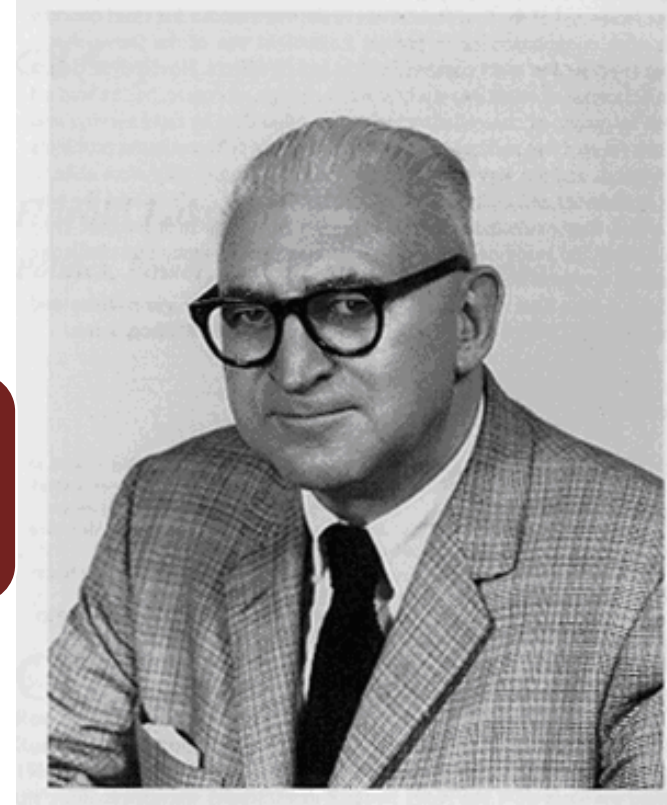
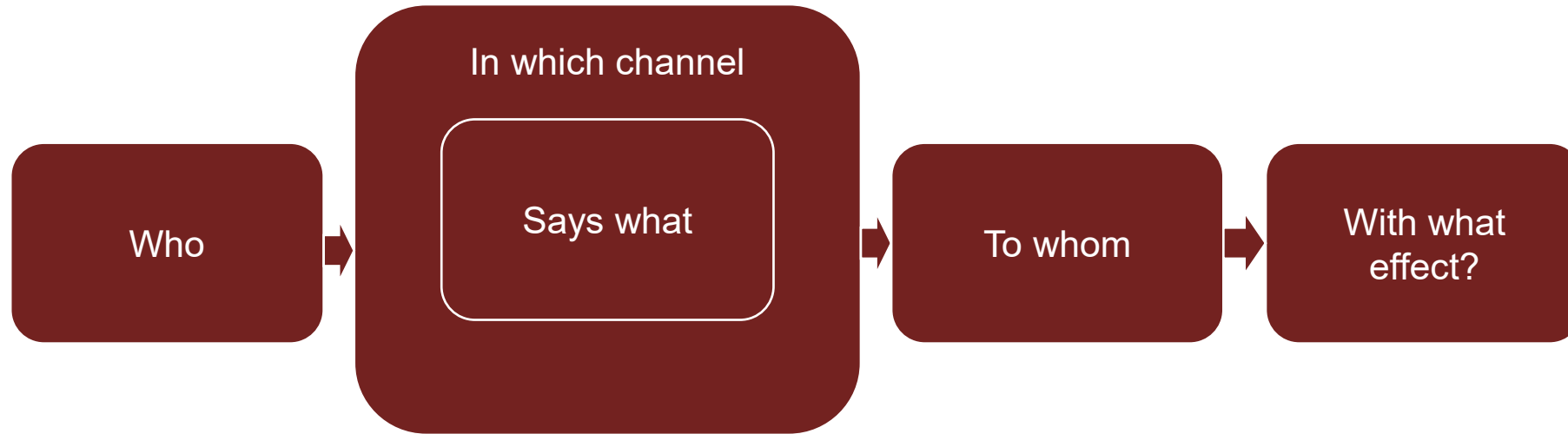
(Sylvia IV, 2024)

- 200 – 100 thousand years ago: The first talking human
(based on genetical examinations)
→ Start of the *talking era* and the ability to share and receive other's signs
- Around 3500 B.CE (5.2 thousand years ago)
→ Start of the *manuscript era* and the overcoming of time and space limitations
along with the rise of agriculture, it became more relevant to take and share notes and plan, at least for some literate elites
- 1450 – 1850: The *print era* and the start of mass-accessible mediated communication:
Overcoming of time and space limitations for recipients
- 1850- 1990: Audio-visual era and the start of mass communication: Overcoming of time and space limitations for senders
- 1990 on: The digital era: Overcoming dissemination and access barriers
- And now? Start of the AI-generated era?

Communication is a process:

The Lasswell formula

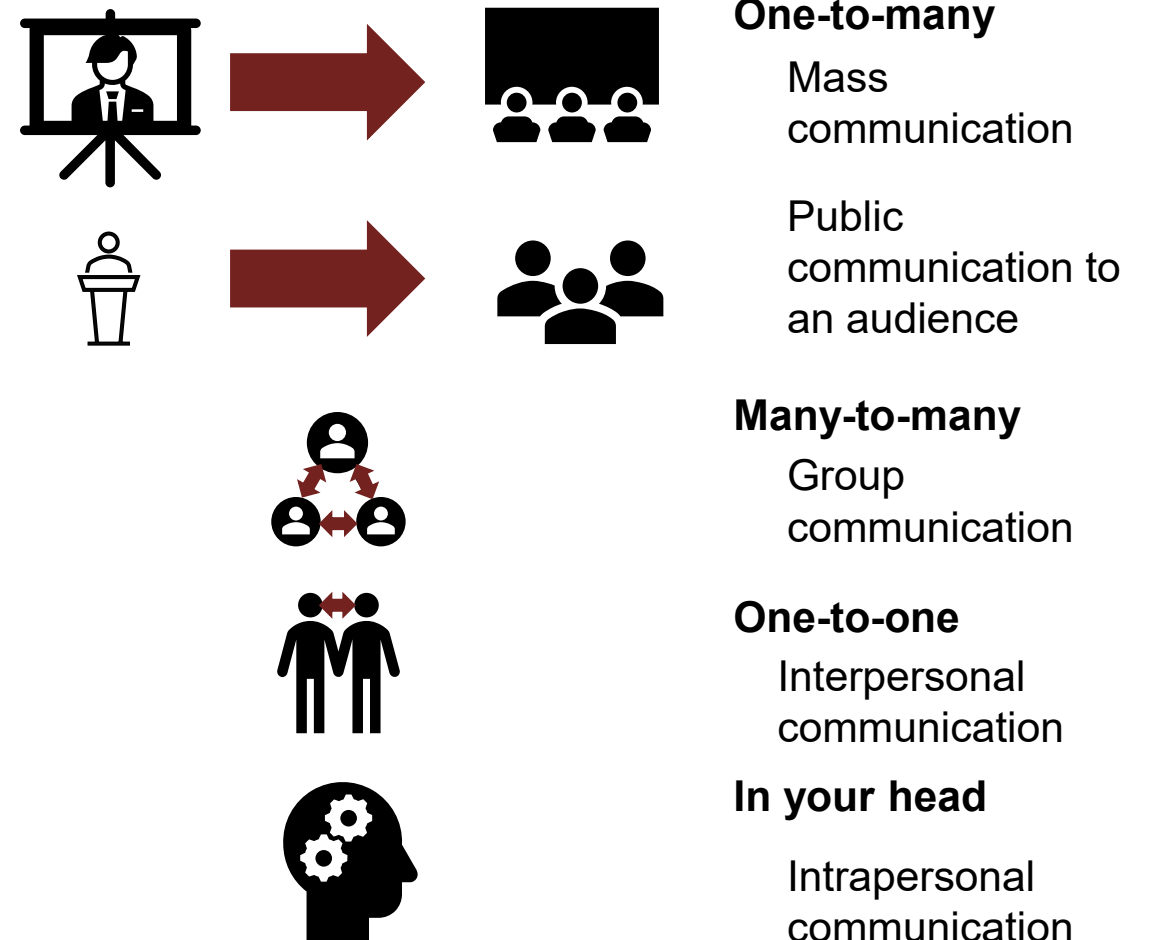
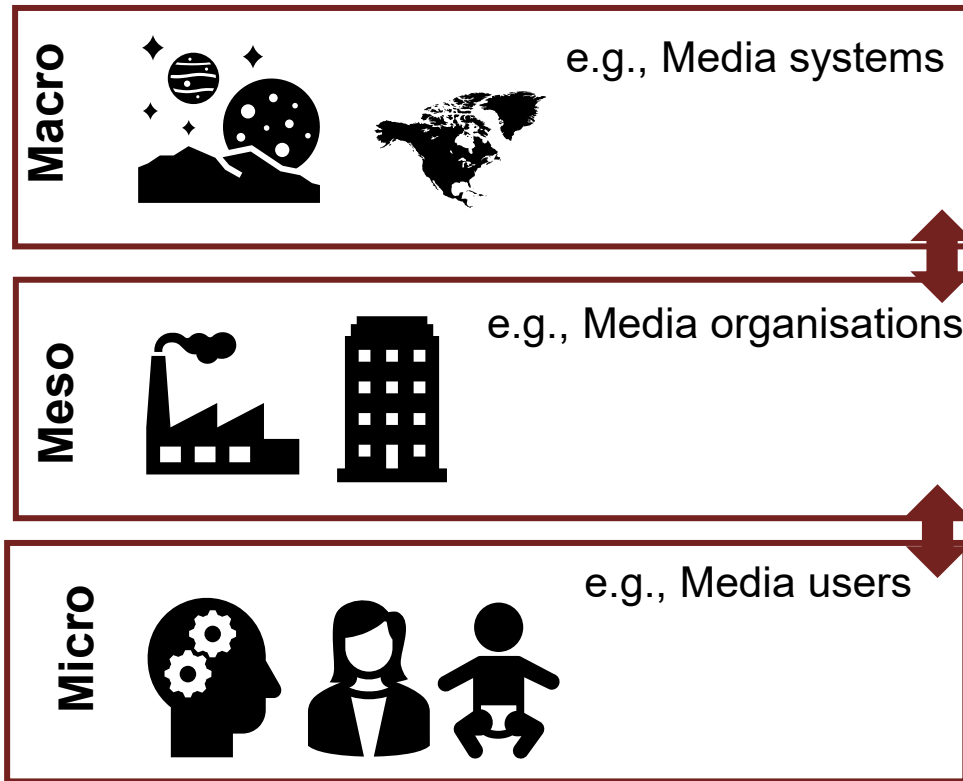
(Lasswell, 1948)



Harold D. Lasswell, by
Martap95 CC BY-SA 4.0,
<https://commons.wikimedia.org/w/index.php?curid=46469016>

Multiple levels of influence and multiple modes of communicating

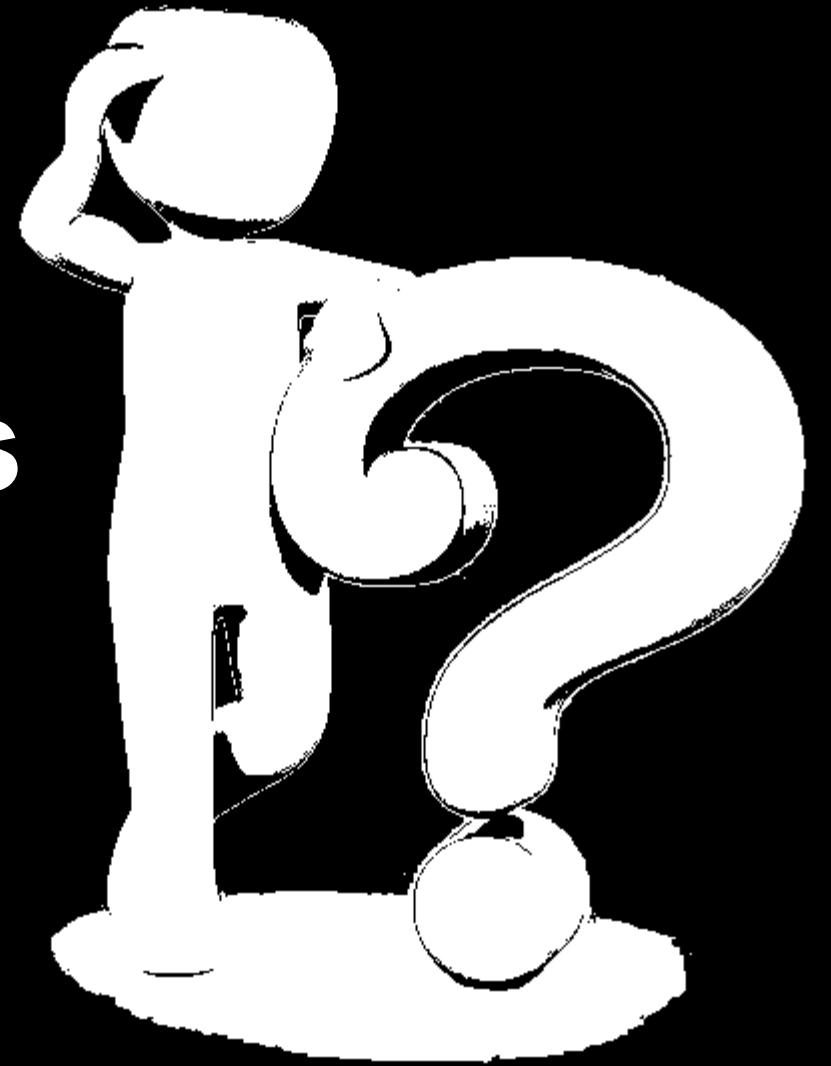
(Sylvia IV, 2024)



**Understanding
communication is
complex!**



**What are your questions
so far?**



**Let's start at the
beginning:
Message production**

How do we speak?

(Griffin & Ferreira, 2006; Berger, 2010)

- Speaking is not easy – it takes up to five times longer to generate a word compared to recognizing it
- Speaking always requires intention (aka working memory capacity & system II thinking)
- Word recognition becomes often automatic
- Writing is also typically intentional
- Doesn't mean it's always a good idea!



Speaking includes different steps:

(Griffin & Ferreira, 2006; Berger, 2010)

1. *Conceptualization* stage: Message is composed out of abstract semantic and pragmatic features → interface between thinking and language production
2. *Formulation* stage: propositions and concepts are represented as lemmas or lexical entries. The lemmas are assigned a functional role, but they must be processed and assigned a positional order afterwards.
3. *Phonological encoding* stage: After the position is clear, the lemmas are encoded in actual grammatical structures
4. *Actual articulation*: the necessary muscles are activated and speaking occurs



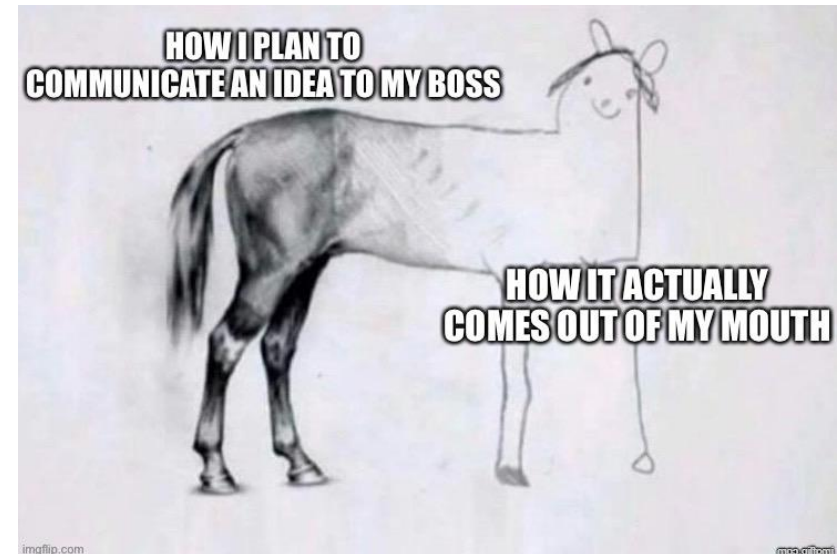
Still debated: Number of encoding stages & feedback loops

Speaking is exhausting (Berger, 2010)

- A lot of speech acts are shortened and contextual to save energy
- It's estimated that around 70% of speech acts are formulaic
- This can – of course – go wrong, speakers might deliver too much or too little information for the audience to get the message
- “The capacity to generate fluent and comprehensible speech by itself does not, ipso facto, guarantee communicative success in social transactions”. (p.114)



<https://www.bbc.com/news/newsbeat-45633362>



<https://imgflip.com/i/4fdtov>

Language happens in social contexts

Berger, 2010)

- Central for successful communication: a common ground in the respective context
 - Shared beliefs
 - Shared assumptions
 - Shared knowledge
- Good communicators estimate this correctly and adapt to it!
- Common ground emerges from two sources:
 - Communal common ground (= community/ culture based)
 - Personal common ground (= participants shared experiences)
 - The better, we get to know each other, the more we can use personal common ground
 - When we get to know someone – we typically try to establish such a shared common ground by asking for preferences, biographical information etc. → Uncertainty reduction

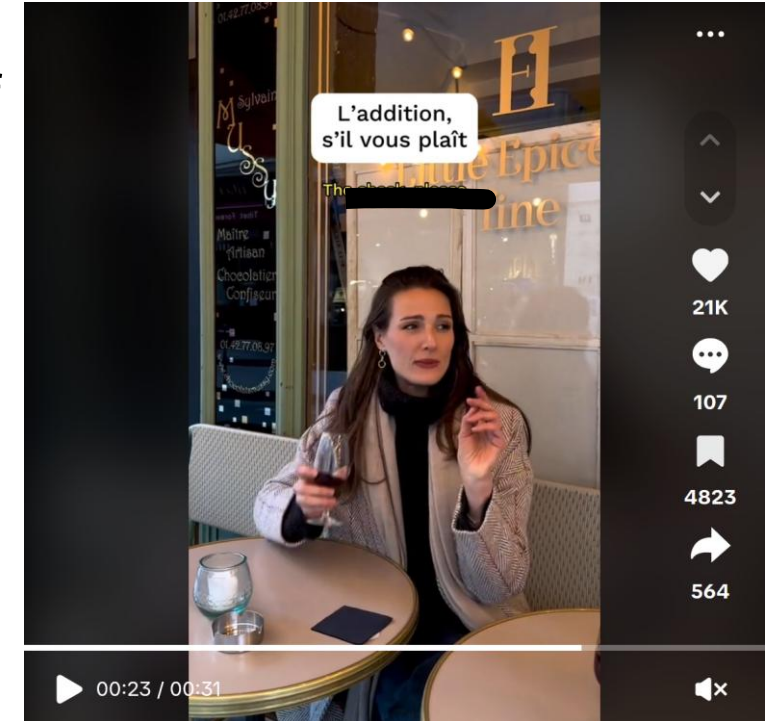


<https://www.someecards.com/usercards/viewcard/MjAxNC1jMDg1MzI1NGViZWl0Y2Rm/>

We talk because we want something

(Berger, 2010)

- Language production is goal-directed
- Effective communication is also dependent on our understanding of others' goals
- Verbal communication is – to a certain extent - cooperative act
 - Ideally: Making truthful, understandable, relevant and parsimonious comments
 - Respectively: Being perceived as making such statements
- Schemes help: With contextual knowledge it is easier to know what the other person wants
- To get what we want, we typically rely on (more or less concrete) plans – for example making an offer to get something



<https://www.tiktok.com/@hellofrenchnyc/video/7210417246647176490>

How does generative AI produce texts?

The transformer

(<https://www.ibm.com/think/topics/transformer-model>; Devlin et al., 2019; Vaswani et al., 2017)

- In 2017, a group of Google engineers, together with Canadian researchers, proposed a neural network with purely attention-based mechanisms – **the transformer**
- Replaced the most common layers in deep-learning architectures (recurrent & convolutional neural networks) with multi-headed self-attention layers
- Self-attention calculates weights for each word in a sentence while accounting for the other words in that sentence (! Human attention = focus, algorithmic attention = context !)
- Basis for current Bidirectional Encoder Representations (*BERT*, encoder) and Generative Pre-trained Transformers (*GPT*, generator).

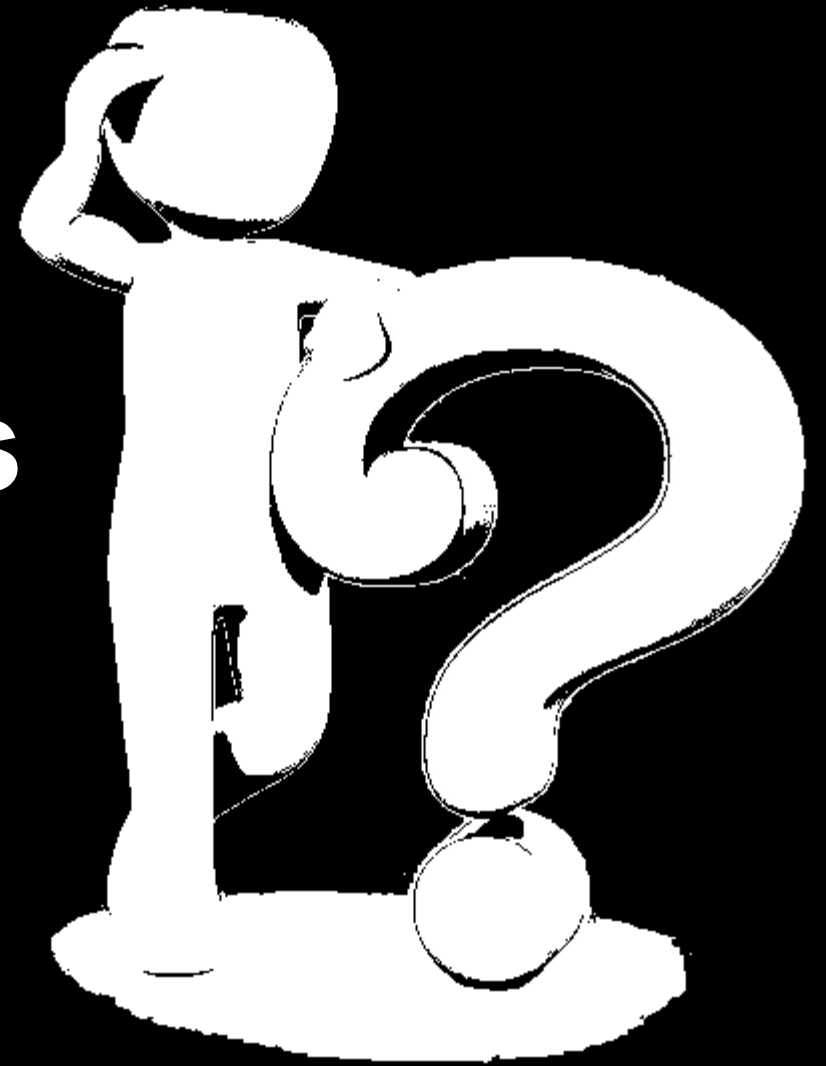
Attention Is All You Need

 https://www.linkedin.com/in/ashish-vaswani-99892181/ Ashish Vaswani* Google Brain avaswani@google.com	 https://www.linkedin.com/in/noam-shazeer-3b27288/ Noam Shazeer* Google Brain noam@google.com	 https://scholar.google.com/citations?user=q2YXPSgAAAAJ&hl=en Niki Parmar* Google Research nikip@google.com	 https://www.linkedin.com/in/jakob-uszkoreit-b238b51/ Jakob Uszkoreit* Google Research usz@google.com
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Human versus Gen-AI: Communication production – what do you think?

Dimension	Human	AI
Intentionality	++	
Stages	Conceptualisation, formulation, phonological encoding, articulation	
Shortened	++	
Common ground	Shared beliefs, assumptions, knowledge, personal or cultural	
Goal directed	Optimized for mutual understanding	

**What are your questions
so far?**



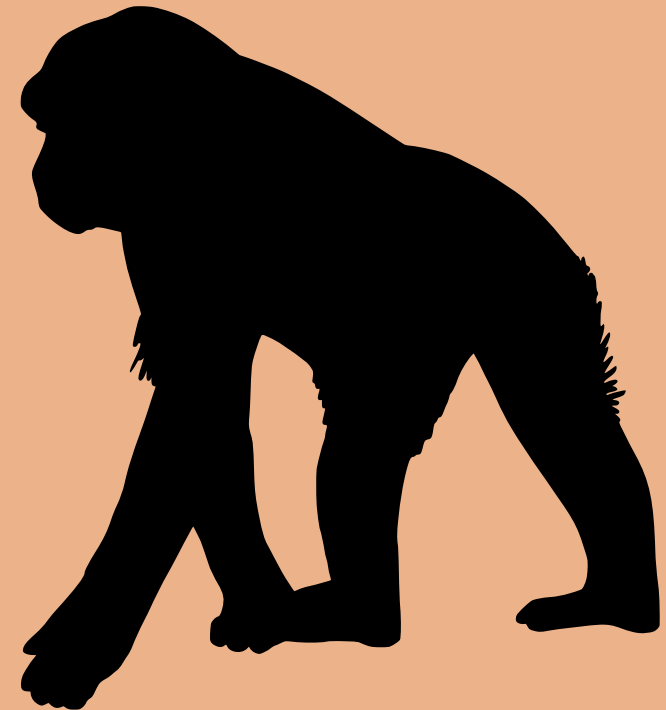
Then its time for a break!



**How do we process
communication?**

What do you think?

What is the necessary precondition for processing messages?



Perception

(Bench et al., 1993; Roskos-Ewoldsen et al. , 2010)

- Perception requires *attention*
- Attention = the process making information available for analysis in the cognitive system (→ Encoding)
- Attention is always selective → some aspects of our environment are processed (those that are deemed relevant), while others are ignored
- Selective attention is not fully under our control

So called *Stroop* effect – test your attention span yourself

<https://www.focusaur.com/pages/stroop-effect-test>

BLUE

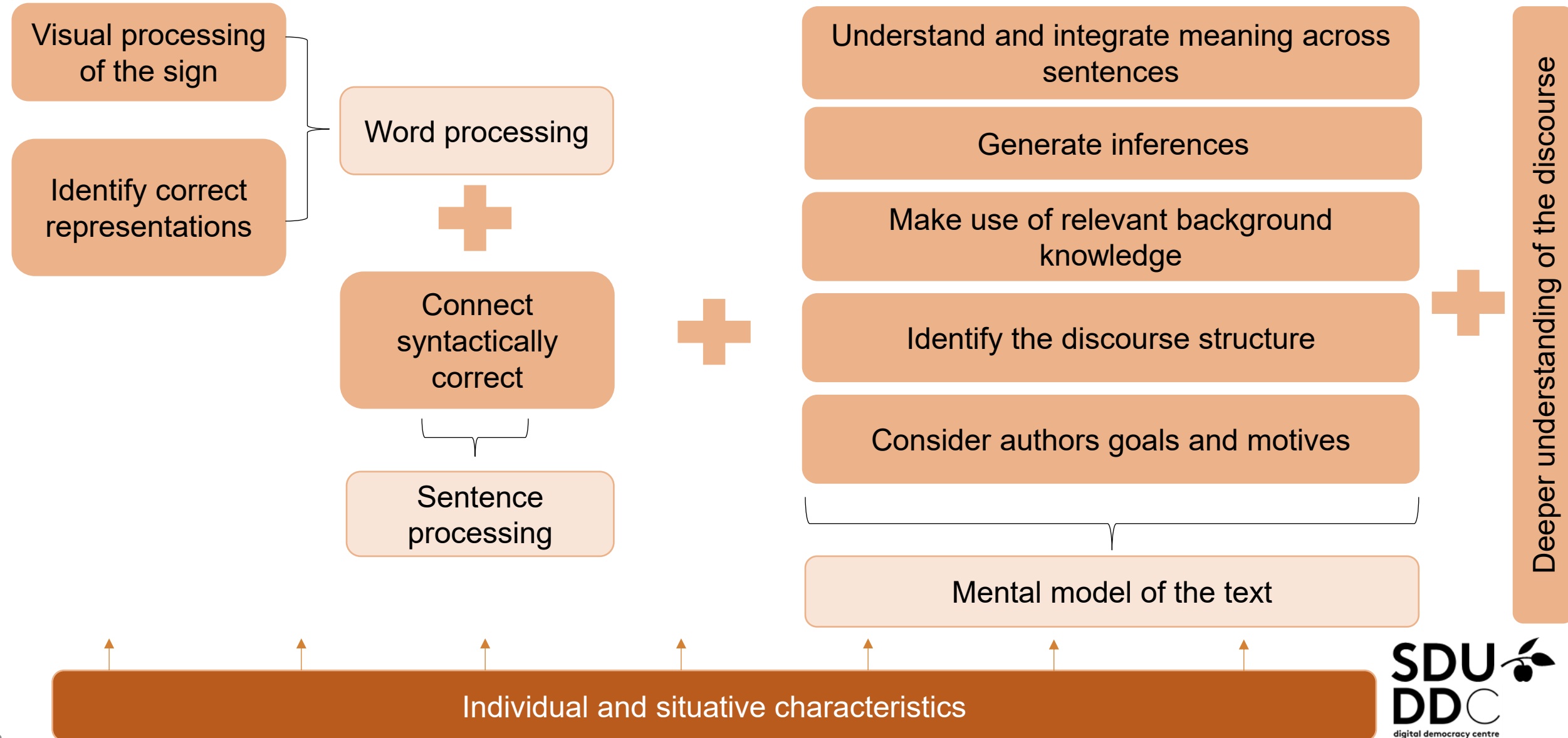
YELLOW

RED

PINK

Text comprehension

(Kendou & O'Brien, 2017)



The limited capacity model of motivated mediated message processing (LC4MP)

(Lang, 2000; Klimmt & Rieger, 2021)



Central task: Select what is encoded from a media stimulus

- Automatic (unintentional) → relevant OR novel/ intense/ surprising
- Controlled (intentional)

Associative

- Content is associatively related to similar content
- Or to content that is temporarily contingent
- Or to positive/ negative valence
- E.g., biographic resonance theory

Getting the content back from the memory

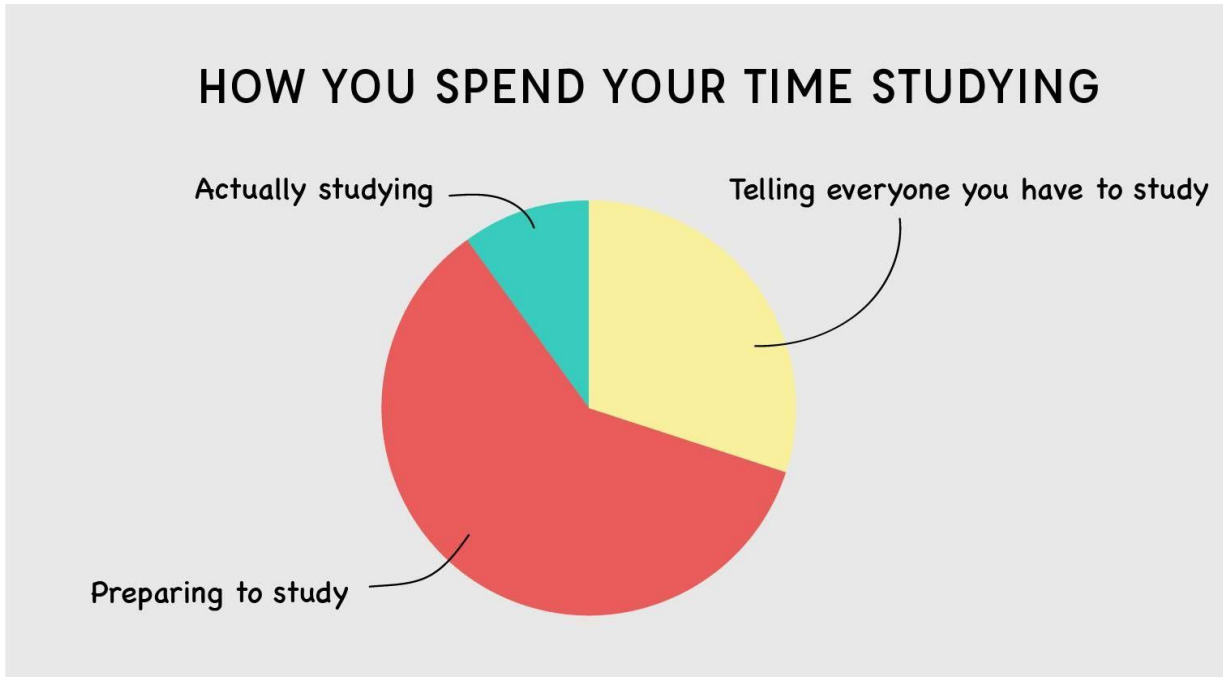
- Easier, when there are more associations
- Active processing during media consumption → interpretation of messages



Annie Lang via <https://cogs.indiana.edu/staff/faculty/profile.php?faculty=an>

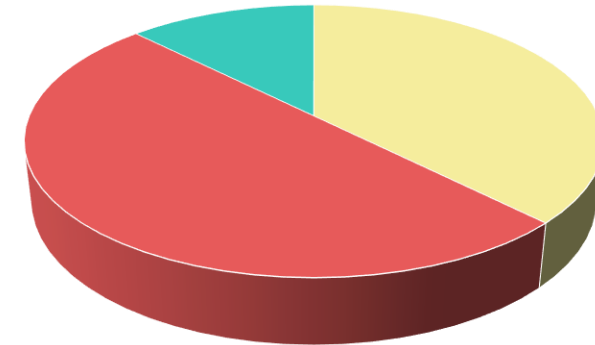
During media consumption ***all*** processes require resources

We all have been there...



<https://www.pinterest.com/learn2earnwebsite/studyingexam-memes/>

In terms of the LC4MP

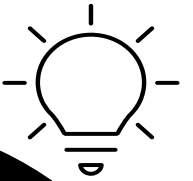


- Resources available
- Resources required
- Resources allocated

Message comprehension

(Kendou & O'Brien, 2017; Roskos-Ewoldsen & Roskos-Ewoldson, 2010)

1. Character perception: We understand mediated messages, because we form a – more or less complex – mental model of the characters driving the storyline
2. We activate concepts following an inner landscape & through iteration (→ *Construction Integration Model*)
 - Concepts that are relevant in the current sentence are activated
 - Concepts from the last sentence should be still activated
 - Concepts can be retrieved if they are relevant to understand the current sentence
 - World knowledge can be retrieved
3. Once concepts are activated, they spread through the neural network
4. Through iteration and feature selection, only relevant aspects are kept



Thema- Rhema
Structure of good
writing!

Sentence 2 should refer
to the concept of
Sentence 1 – actively
activate world – or prior
knowledge

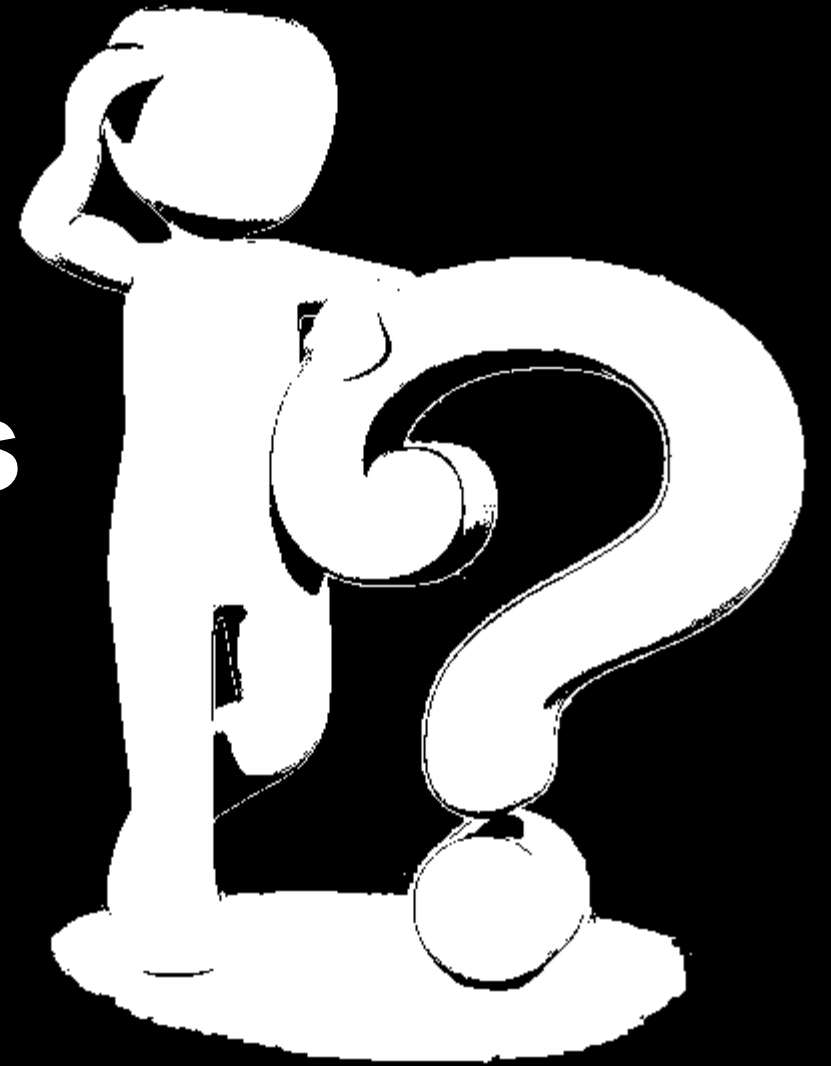
AI as universal translator and voice storage?



What are the risks/
potentials of artificially
conserved/ generated
voice?

**How do you protect your family members
and/ or future company?**

**What are your questions
so far?**



How contexts & computers shape Communication

Communication changes between contexts

(Soliz & Giles, 2010; Giles, 2016)

- Humans adapt their communication to their communication partners
(→ *Communication accommodation theory, CAT*)
 - Convergence: we adjust our motion, haptics, proxemics (closeness), para-language (speaking style), etc. to our interaction partners to reduce social distance (→ e.g., ingroup members)
 - Divergence: we choose motions, haptics, etc., to accentuate the difference between us and others (→ e.g., outgroup members)
 - Within one talk, we often combine both strategies

Adaptation can be

- Appropriate → great
- under accommodated → selfish
- Overaccommodated → patronizing



Howard Giles
<https://econtent.hogrefe.com/doi/10.1026/1617-6383.16.1.17>

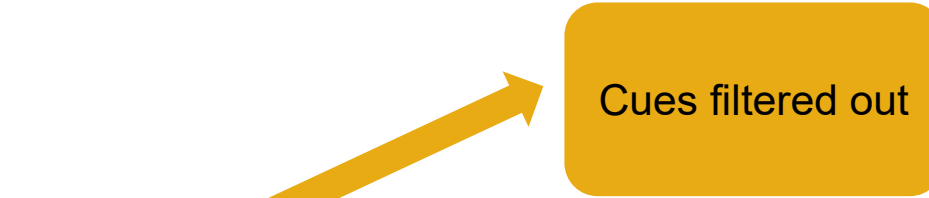
Consider these people just had their 1st date – who is going to meet again?



Computer-mediated communication (Walther et al., 2010)

Can differ from face-to-face communication regarding the

- Synchronicity
- Permanency
- Anonymity
- Less cues (e.g., lack of nonverbal communication)
- Other cues (e.g., emojis)

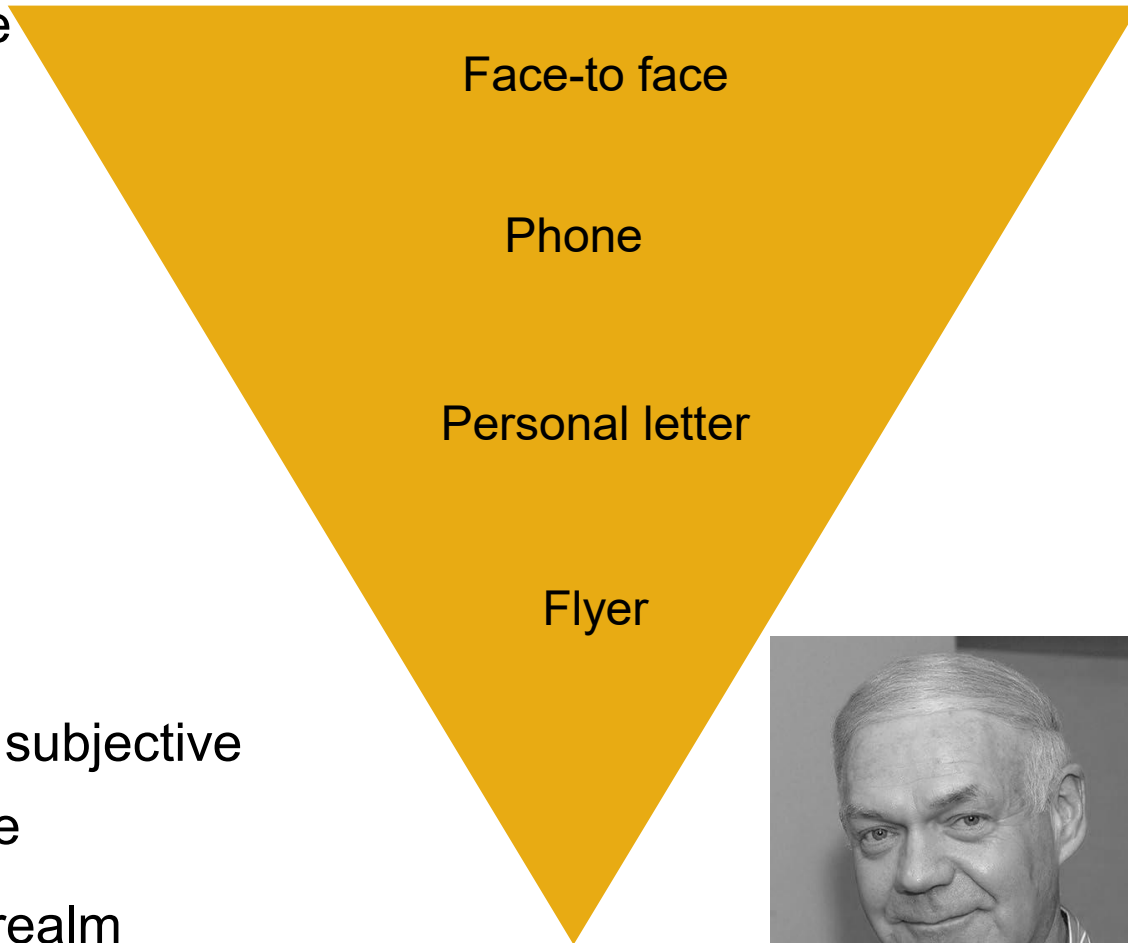


- communication mediated by technology filters out many cues found in face-to-face interaction
- different media filter out or transmit different cues
- substituting technology-mediated for face-to-face communication will result in predictable changes → typically things get worse
- E.g., *Media richness theory*

Media richness theory

(Draft & Lengel, 1984a, b; Ishii et al., 2019)

- Basic idea: Match the richness of the chosen communication medium to the significance of reducing uncertainty and establishing equivocality
- Richness = immediate feedback, multiple cues, language variety, personal focus
- E.g., arranging a meeting vs. firing someone
- Later formulations also accounted for subjective richness and norms around media use
- Relationships less clear in the digital realm



Richard Draft by Vanderbilt University's media relations unit., <https://commons.wikimedia.org/w/index.php?curid=6597367>



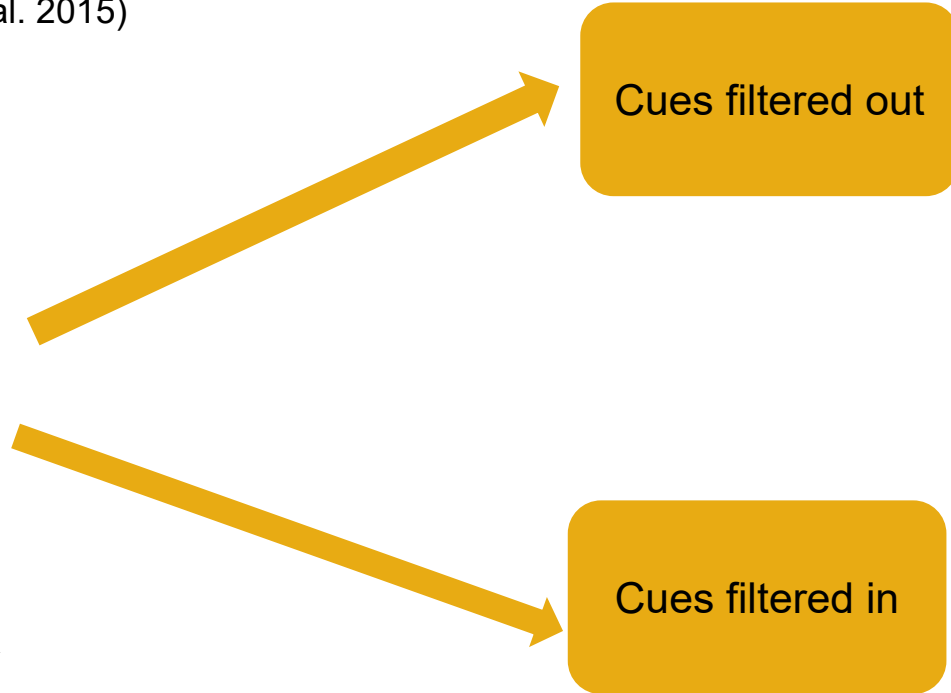
<https://www.amazon.de/stores/author/B07GBKGMDC>

Computer-mediated communication

(Walther et al. 2010; Walther et al. 2015)

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<https://rebootingsocialmedia.org/person/joseph-b-walther/>

- *Social Information Processing Theory, Hyperpersonal model of computer-mediated communication*
- Users exploit communicative content and style → different cues are used to fulfil specific functions and, if cues are missing, other cues replace them (#emojis...)
- We can present ourselves more intentionally and are perceived to be more intentional

The Social Identity Deindividuation Effects Model (SIDE)

(Cialdini et al, 1990; Kallgren et al., 2000; Postmes, Spears, & Lea, 1998)

- Based on Social Identity/ Self-categorization theory
- In anonymous digital contexts, our personal identity is less salient
- Instead: Stronger orientation towards the norms in the respective digital context
- If the norms are hostile: More hostile behaviour
- If the norms are constructive: More constructive behaviour
- Norms emerge injunctive (what you should do) and descriptive (what others do) → *Focus Theory of Normative Conduct*



Tom Postmes
https://research.rug.nl/files-asset/145913516/IMG_1949.jpg?w=320&f=webp



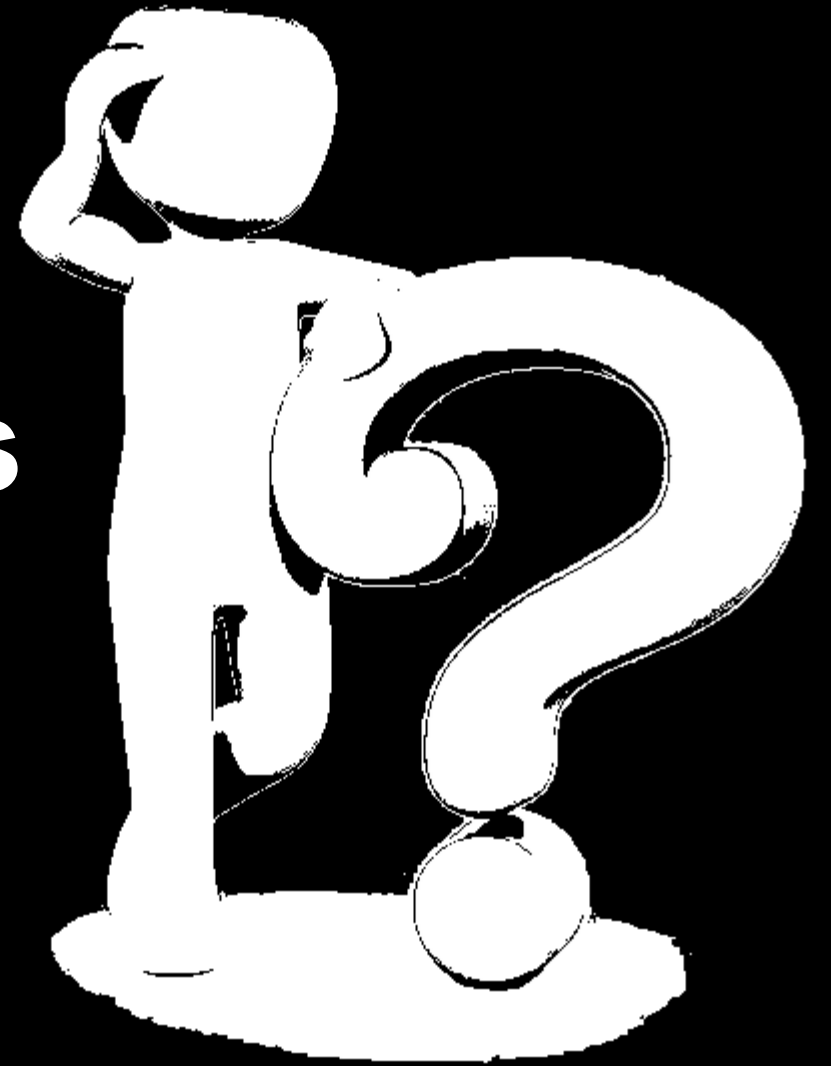
[ps://research.rug.nl/en/persons/john-s-spears](https://research.rug.nl/en/persons/john-s-spears)



<https://martinlea.com/>

**What do you think: How
will interacting with GenAI
empowered chatbots
shape our
communication?**

**What are your questions
so far?**



Then I hope, by now you

- ...have an initial understanding of communication
- ...have examined the production and processing of communication
- ...have gotten a glimpse into the complex processes behind reading
- ...explored how cultures and computers shape communication

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AI 507: Artificial Intelligence and Society



Have a great rest of the day